

EFFECT OF COMPATIBILIZER IN IMPROVING MECHANICAL PROPERTIES OF PLA (POLYLACTIC ACID) OR POLYLACTIC ACID BLENDS, BASED GREEN HYBRID COMPOSITES



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**MECHANICAL ENGINEERING
BACHELOR OF TECHNOLOGY
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**EFFECT OF COMPATIBILIZER IN IMPROVING
MECHANICAL PROPERTIES OF PLA (POLYLACTIC
ACID) OR POLYLACTIC ACID BLENDS, BASED GREEN
HYBRID COMPOSITIES IN THE APPLICATION OF
WIND TURBINE BLADE**

Thesis submitted to

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for the award of the degree

of

Bachelor's of Technology

by

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**MECHANICAL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY
MAY - 2024**

APPROVAL SHEET

This report entitled “**Effect of compatibilizer in improving mechanical properties of PLA (poly lactic acid) or poly lactic acid blends, based green hybrid composites**” by Biprajit Majumder, Rajesh Sarkar, Ankita Debbarma and Joydeep Debbarma is approved for the degree of Bachelor of Technology.

Examiners

Supervisor (s)

Chairman

Date: _____

Place: _____

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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CERTIFICATE

It is certified that the work contained in the thesis titled “, Effect of compatibilizer in improving mechanical properties of PLA (poly lactic acid) or poly lactic acid blends, based green hybrid composites” by “Biprajit Majumder,” “Rajesh Sarkar,” “Joydeep Debbarma” and “Ankita Debbarma” has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Signature of Supervisor
Prof. (Dr.) Pinku Debnath
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May, 2024

PREFACE

The advent of renewable energy technologies has sparked a revolution in the quest for sustainable solutions to power our world. Among these, wind energy stands out as a beacon of promise, offering clean and abundant power to fuel our future. Central to the efficiency and reliability of wind turbines are their blades, which must withstand formidable forces while maintaining structural integrity over time.

In the pursuit of enhancing the mechanical properties of materials used in wind turbine blades, the role of compatibilizers in polylactic acid (PLA) and PLA blends cannot be overstated. Compatibilizers serve as catalysts for synergy, bridging the gap between disparate materials to create green hybrid composites that exhibit superior mechanical performance.

This work delves into the intricate interplay between compatibilizers and PLA-based materials, unravelling the mechanisms by which they bolster mechanical strength, durability, and resilience. Through meticulous experimentation and analysis, we aim to shed light on the transformative potential of compatibilizers in advancing the application of PLA blends in wind turbine blade construction.

As we embark on this journey of exploration and discovery, we invite readers to join us in unravelling the complexities of materials science and renewable energy innovation. Together, let us envision a future where sustainable solutions reign supreme, and where the humble wind turbine blade stands tall as a testament to human ingenuity and environmental stewardship.

ACKNOWLEDGEMENT

We would like to express our thankful gratitude to our project supervisor Dr. Pinku Debnath, Assistant Professor, Department of Mechanical Engineering, NIT Agartala, for his cooperation, suggestion, active support, and immense help.

We are extremely fortunate to be involved in such an exciting and challenging project. It gave us an opportunity to work in a new environment of fluent. This project has increased our thinking and understanding capability.

We would like to express our thanks to all our friends, all staffs and faculty members of the department for making our stay in NIT Agartala a pleasant and memorable experience. We would also like to convey our heartiest gratitude to our parents for their unconditional love and support.

Yours faithfully,

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ABSTRACT

This study investigates the effect of compatibilizers on enhancing the mechanical properties of polylactic acid (PLA) or polylactic acid blends-based green hybrid composites for applications in wind turbine blades. With the increasing demand for renewable energy sources, there is a growing interest in utilizing biodegradable and sustainable materials in wind energy systems. PLA, derived from renewable resources such as corn starch or sugarcane, offers promising characteristics such as biodegradability, biocompatibility, and low carbon footprint, making it an attractive candidate for composite materials in wind turbine blades.

However, PLA-based composites often exhibit limitations in mechanical properties such as tensile strength, flexural strength, and impact resistance, which are critical for the structural integrity and performance of wind turbine blades. To address these challenges, compatibilizers are introduced to improve the interfacial adhesion and compatibility between PLA and reinforcement materials such as natural fibres or biodegradable additives.

This review explores recent advancements in the use of compatibilizers to enhance the mechanical properties of PLA-based composites for wind turbine blade applications. Various types of compatibilizers, including maleic anhydride-grafted PLA, reactive coupling agents, and bio-based compatibilizers, are evaluated for their effectiveness in improving interfacial bonding and mechanical performance.

Furthermore, the influence of compatibilizer type, concentration, processing conditions, and composite formulation parameters on the mechanical properties of PLA-based composites is discussed. The synergistic effects of compatibilizers in enhancing tensile strength, flexural strength, impact resistance, and fatigue behaviour are examined through experimental studies and numerical simulations.

Overall, this review provides insights into the role of compatibilizers in optimizing the mechanical properties of PLA-based green hybrid composites for wind turbine blade applications, highlighting their potential to contribute to the advancement of sustainable and eco-friendly materials in renewable energy systems.

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CHAPTER – 1

INTRODUCTION

Polylactic acid (PLA) and its blends have garnered significant attention as sustainable alternatives to traditional petroleum-based polymers in various industrial applications. Their biodegradability, renewability, and low environmental impact make them attractive candidates for eco-friendly materials. Among the potential applications, the use of PLA-based composites in wind turbine blades offers a compelling avenue for reducing the environmental footprint of renewable energy infrastructure. However, the mechanical properties of PLA, such as its tensile strength, flexural strength, and impact resistance, often fall short of the requirements for demanding applications like wind turbine blades, which necessitate materials with superior strength and durability.

Incorporating reinforcing fibres, such as natural fibres or glass fibres, into PLA matrices has been a common strategy to improve mechanical properties. However, achieving a strong interfacial adhesion between the hydrophilic PLA matrix and hydrophobic reinforcing fibres remains a challenge, leading to suboptimal mechanical performance and limited scalability of PLA-based composites in high-performance applications.

One promising approach to address this challenge is the use of compatibilizers, which are additives designed to enhance the compatibility between polymer matrices and reinforcing fibres. By promoting adhesion at the interface, compatibilizers can improve the dispersion of reinforcing fibres within the matrix and facilitate stress transfer, thereby enhancing the mechanical properties of the composite material.

In the context of wind turbine blade applications, where structural integrity and fatigue resistance are paramount, the effectiveness of compatibilizers in PLA-based green hybrid composites becomes particularly crucial. This study aims to investigate the role of compatibilizers in improving the mechanical properties of PLA-based composites tailored for wind turbine blade applications. Through systematic experimentation and analysis, this research seeks to elucidate the mechanisms underlying the enhancement of mechanical properties and provide insights into the design and optimization of sustainable composite materials for renewable energy infrastructure.

EXPERIMENTAL

The experimental work encompasses the extraction and treatment of fibers and the development and characterization of the composites. In this section, a detailed discussion of the raw materials and procedure followed for the development of composites are provided. Moreover, the characterizations and investigation of the fibers are provided in this section.

MATERIALS

The composites are developed by taking PLA (poly lactic acid) as matrices with the incorporation of jute and sisal fibers as reinforcement. The typical properties of the Poly lactic acid. C o c o n u t f i b e r and Sisal (Agave sisalana) are extracted from the corresponding base raw materials. The properties of the jute and sisal fibers are furnished, give the images of the raw materials used for the development of the fiber-reinforced polymeric composites.

CHARACTERIZATION OF FIBERS

The surface morphology of the jute and coconut fibers are examined by using SEM techniques. The sample is fixed over a conductive adhesive tape which is coated with gold through an ion sputter, and observations are made by using a microscope of model JEOL JSM 5310 which is operated at an applied voltage of 15 kV.

POLYLACTIC ACID (PLA)

Polylactic acid (PLA) is a renewable thermoplastic polymer matrix and an aliphatic polyester made from lactic acid and has been a key component in producing biodegradable products, making it one of the most promising biopolymers used in the world. Generally, PLA is produced from agricultural source materials that have been fermented to produce lactic acid. The cyclic dimer (lactide) of lactic acid is polymerized through the combination of oligomerization and cyclic dilactone processes. PLA has been proven to have exceptional properties through thorough studies since 1990 and was estimated to increase production capacities. Because PLA is completely biodegradable, it has been commercialized for applications that need biodegradability, such as disposable cups and plates, packaging materials, disposable cutlery, and plastic bags.

ADVANTAGES AND DISADVANTAGES OF POLYLACTIC ACID



Figure 1: PLA

Advantages:

1. Biodegradability: PLA is derived from renewable resources and is biodegradable, making it environmentally friendly.
2. Versatility: It can be processed into various forms including films, fibres, and 3D printing filaments.
3. Low Toxicity: Generally considered safe for use in food packaging and medical implants.
4. Renewable Source: PLA is made from renewable resources like corn starch or sugarcane, reducing reliance on finite fossil fuels.

Disadvantages:

1. Limited Heat Resistance: PLA has a relatively low melting point compared to some other plastics, limiting its use in high-temperature applications.
2. Brittleness: PLA can be relatively brittle, especially in thin or intricate designs.
3. Moisture Sensitivity: It can degrade when exposed to moisture, affecting its mechanical properties over time.
4. Slow Decomposition: PLA requires specific conditions (e.g., industrial composting facilities) to decompose, which may not be readily available in all environments.

PHYSICAL PROPERTIES OF PLA

In general, dioxane, acetonitrile, chloroform, methylene chloride, 1,1,2-trichloroethane, and dichloroacetic acid can dissolve PLA and most of its derivatives. At low temperatures, PLA products can be partially dissolved in ethyl benzene, toluene, acetone, and tetrahydrofuran (THF). However, at boiling temperatures, they can be fully dissolved in the solvents. Water, alcohol, and aliphatic hydrocarbons such as hexane and heptane, which can be used as antisolvents, are incompatible with PLA products. In acetone, ethyl acetate, or THF solvents, PLA with high crystallinity and molecular weight is insoluble, and only polymers with lower molecular weights can be dissolved.

Polymers can be semi crystalline or amorphous. Semi crystalline polymers have regular repeating units that cause the chains to fold into crystallites, which are dense regions. The PLA can be either amorphous or semi crystalline at room temperature, depending on the molecular weight and stereochemistry structure. It has been reported that the melting enthalpy for 100% crystallinity PLA is 93–148 J/g. PLA is made of pure L-lactic and D-lactic isomers, generating poly-L-lactic acid (PLLA) and poly-D-lactic acid (PDLA) homopolymers. Additionally, Poly-D,L-lactic acid (PDLLA) copolymer is produced by using a racemic combination of L- and D-monomers. It was reported that PDLLA is an amorphous polymer with no melting point, while PLLA is a semi-crystalline polymer. Because of the existence of crystalline areas, the degradation rate of PLLA is considerably slower than that of PDLLA.

The amorphous and crystalline densities of PLLA have been stated to be 1.248 g/mL and 1.290 g/mL, respectively. L-lactide has a density of 1.36 g/cm³, D-lactide has a density of 1.33 g/cm³, crystalline polylactide has a density of 1.36 g/cm³, and amorphous polylactide has a density of 1.25 g/cm³. The viscosity of polylactic acid solutions increases as the concentration increases but decreases as the temperature rises.

The distribution functions of crystallites in PLA were determined as a function of spiral angle by Kobayashi et al. They also verified the helical structure of PLLA and PDLA and showed that PLLA had developed tremendous optical activity in the solid- state. In the lower UV-C range (190–220 nm), PLA has almost no UV transmission. The amount of UV light transmitted by PLA is substantially increased at 225 nm. The overall properties of PLA can be regulated by using special catalysts that have isotactic and syndiotactic content with different enantiomeric units. Below the table summarizes the general properties of commercial amorphous PLA.

MECHANICAL PROPERTIES OF PLA

PLA is a bio-based polymer with the potential to replace petroleum-based plastics due to its stiffness and strength. The mechanical characteristics of PLA may range from elastic soft to stiff, high-strength materials, depending on numerous factors such as polymer structure, material formulation, orientation, crystallinity, and molecular weight as shown in below table. When higher mechanical properties are needed, semi crystalline PLA is favored over amorphous PLA. Farah et al. discovered that generally, PLA has a tensile modulus of 3–4 GPa, a tensile strength of 50–70 MPa, a flexural strength of 90–100 MPa, and an elongation at break of around 1–4%.

Unlike its thermal properties, PLA's mechanical properties and crystallization behavior are highly dependent on Mw and the stereochemical make-up backbone. For instance, when the Mw is increased from 50 to 100 kDa, the tensile modulus of PLLA increases by a factor of two, and tensile strengths of 15.5, 80, and 150 MPa were obtained for varying the Mw from 50 to 150 to 200 kDa, respectively.

Tensile properties—tensile strength (σ , in MPa), tensile modulus (E, in GPa), ultimate strain (ϵ , in percent), and polymer density (ρ , in g/cm³)—are the mechanical properties of PLA that have been studied the most in relation to a collection of biopolymers. Tensile properties are clearly best for the densest recorded polymers, especially poly (glycolic acid) (PGA). PCL, on the other hand, appears to be the most fragile polymer, with a heavy strain at failure. Data on flexural properties were insufficient to make a comparison. However, since flexural and tensile properties are largely associated, the tendencies observed here are likely to be the same as those observed when comparing flexural properties.

PLA tensile strength and modulus ranges are compared to various biopolymers. These characteristics are crucial since they specify the dimensions required for a given mechanical strength or stiffness. PGA and PLLA appear to be the best options in this situation, despite their lack of utility as a composite matrix, while PCL and polyhydroxy butyrate (PHB, σ *8.8 MPa and E*7.8 MPa) are obviously the worst.

In terms of flexural properties, PLA is a promising thermoplastic polymer having a flexural strength of up to 140 MPa and Young's modulus of 5–10 GPa and good optical characteristics. The addition of natural fibers to PLA may enhance its mechanical characteristics. The applicability of natural fibers such as hemp and flax as a reinforcement material was addressed by Loos et al., and the mechanical characteristics of flax and hemp employed in polymer reinforcement were studied. Previously, researchers showed that adding natural fibers to PLA bio composites improves their flexural, tensile, and impact strength.

THERMAL PROPERTIES OF PLA

T_g and T_m are present in semi crystalline PLA, as they are in many thermoplastic polymers. PLA is rubbery above T_g (58 °C), but it becomes a glass below T_g, which can also creep until cooled to its transition temperature of approximately 45 °C, below which it acts as a brittle polymer. PLA's T_g and T_m values are compared to those of other polymers in below figure; PLA has a relatively high T_g and low T_m as compared to other thermoplastics, as seen. PLA's T_g is determined by the molecular weight and optical purity of the polymer. The T_g increases with molecular weight, reaching maximum values of 60.2, 56.4, and 54.6 °C for PLA with 100%, 80%, and 50% l-stereoisomer contents, respectively, at the infinite molecular weight. PLA with a higher l-lactide content has higher T_g values than PLA with the same d-lactide content. Tsuji and Ikada found similar relationships. In addition, PLA characteristics may be adjusted by using isotactic and syndiotactic content catalysts with various enantiomeric units. For instance, PLA with a PLLA concentration of more than 90% is crystalline, while PLA with a lower optical purity is amorphous. With decreasing quantities of PLLA, the T_m and T_g of PLA drop. table displays the glass transition and melting temperatures of various PLA polymers made with different copolymer ratios.

Chapter-2

MATERIALS USED

Natural Fibre:

- Coconut fibre:



Figure: 2 - COCONUT FIBRE

Coconut fibre, also known as coir or coconut coir, is a natural fibre extracted from the husk of coconuts. It is a versatile material with various applications across industries.

- a. Natural and Sustainable: Coconut fibre is an eco-friendly material because it is derived from the renewable resource of coconut husks, which are typically discarded after harvesting the coconut fruit.
- b. Strong and Durable: Despite being lightweight, coconut fibre is remarkably strong and durable, making it suitable for a wide range of applications where strength and resilience are important.
- c. Water and Rot Resistant: Due to its high lignin content, coconut fibre is naturally resistant to water, rot, and saltwater degradation, making it ideal for outdoor and marine applications.
- d. Absorbent: Coconut fibre has excellent moisture absorption properties, making it useful in applications such as soil erosion control, gardening, and horticulture, where moisture retention is crucial.
- e. Insulating: Coconut fibre exhibits good thermal insulation properties, making it suitable for use in products such as mattresses, upholstery padding, and acoustic insulation panels.

- f. Versatile Applications: Coconut fibre finds applications in various industries, including agriculture (as a soil amendment and erosion control material), horticulture (as a growing medium and substrate), automotive (as padding and insulation), construction (as reinforcement in composites), and handicrafts (for making ropes, mats, and other woven products).
- g. Biodegradable: At the end of its lifecycle, coconut fibre is biodegradable, meaning it can decompose naturally without causing harm to the environment.

- **Sisal Fibre:**



Figure: 3 – Sisal Fibre

Sisal fibre, extracted from the leaves of the *Agave sisalana* plant, has a wide range of applications due to its strength, durability, and versatility.

- a. Ropes and Twines: Sisal fibre is renowned for its strength and durability, making it ideal for manufacturing ropes, twines, and cordage. These products are widely used in agriculture, shipping, construction, and various other industries.
- b. Agricultural Applications: Sisal fibre is used in agriculture for baling hay and straw. It is also used to make agricultural twines for tying plants, vines, and trees, providing support and stability.
- c. Carpet and Rug Backing: Sisal fibre is often used as a backing material for carpets and rugs due to its natural strength and resistance to wear and tear. It provides stability and durability to these floor coverings.
- d. Erosion Control: Sisal erosion control blankets and mats are used to stabilize soil on slopes and prevent erosion. The natural fibres help retain moisture and promote vegetation growth while protecting the soil from erosion.

- e. Geotextiles: Sisal fibre is utilized in geotextiles for soil stabilization, drainage, and erosion control applications. Geotextiles made from sisal are environmentally friendly and biodegradable, making them suitable for use in landscaping and civil engineering projects.
- f. Crafts and Artisans: Sisal fibre is popular among artisans and crafters for making various handicrafts, including baskets, hats, bags, and decorative items. Its natural texture and aesthetic appeal make it a preferred choice for eco-friendly and sustainable crafts.
- g. Furniture and Upholstery: Sisal fibre can be woven into fabrics and textiles for upholstery, chair seats, and cushions. Its durability and resistance to wear make it suitable for furniture that receives heavy use.
- h. Paper and Packaging: Sisal fibre is used in the production of specialty papers, including tea bags, banknotes, and currency papers. It is also employed in packaging materials such as corrugated boards and cartons for its strength and tear resistance.

• **COMPATIBILIZER:**

Compatibilizers are used to promote interfacial adhesion between pair of polymers which are otherwise immiscible. They are compatible with one of the phases, these molecules tend to concentrate at the interfaces and stabilize them thus allowing finer dispersion of mutually incompatible pairs.

- a. **Ethylene glycol methacrylate (EGMA)** is a monomer used in the production of various polymers and copolymers. It belongs to the family of methacrylate monomers, which are widely employed in industries such as coatings, adhesives, textiles, and electronics. Ethylene glycol methacrylate is a chemical compound used in various applications, especially in the production of polymers and coatings. It's commonly employed in the synthesis of hydrogels, adhesives, and dental materials due to its ability to form cross-linked networks, which offer desirable properties such as flexibility and toughness. Ethylene glycol methacrylate (EGMA) is a monomer used in the production of various polymers and copolymers. It belongs to the family of methacrylate monomers, which are widely employed in industries such as coatings, adhesives, textiles, and electronics. Here are some key points about ethylene glycol methacrylate:
 - i. Chemical Structure: Ethylene glycol methacrylate has the chemical formula $C_8H_{14}O_3$ and the molecular formula $CH_2=C(CH_3)COOCH_2CH_2OH$. It contains a methacrylate group ($-CH_2=C(CH_3)COO-$) and an ethylene glycol side chain ($-OCH_2CH_2OH$).
 - ii. Monomer Properties: EGMA is a clear, colourless liquid with a characteristic odour. It has a relatively low viscosity, which makes it

easy to handle and process in polymerization reactions. It is soluble in organic solvents such as acetone, ethanol, and chloroform.

- iii. Polymerization: Ethylene glycol methacrylate can undergo free radical polymerization to form homopolymers or copolymers with other monomers. In copolymerization, EGMA is often used as a comonomer with other acrylic or methacrylic monomers to impart specific properties to the resulting polymer, such as flexibility, adhesion, or hydrophilicity.
- iv. Applications: EGMA-based polymers and copolymers have a wide range of applications across different industries:
- v. Coatings and Paints: EGMA copolymers are used in architectural coatings, automotive paints, and industrial coatings due to their excellent adhesion, weather resistance, and durability.
- vi. Adhesives: EGMA-containing polymers are used in pressure-sensitive adhesives, sealants, and construction adhesives due to their high tackiness, flexibility, and bond strength.
- vii. Textiles: EGMA-based polymers are used as coatings for textiles to improve their water repellence, stain resistance, and durability.
- viii. Electronics: EGMA copolymers are used in the production of photoresists and electronic packaging materials due to their high resolution, thermal stability, and adhesion to substrates.
- ix. Biomedical Applications: EGMA-containing polymers are investigated for biomedical applications such as drug delivery systems, tissue engineering scaffolds, and biomedical coatings due to their biocompatibility and tunable properties.
- x. Safety Considerations: Like other methacrylate monomers, ethylene glycol methacrylate should be handled with care due to its potential to cause skin and eye irritation. Proper safety precautions, including the use of personal protective equipment and adequate ventilation, should be observed during handling and processing.

Chapter-3

PEPARATION OF COMPOSITE: First of all we take polylactic acid , natural fiber and compitibilizer and then we mix it up. After adding Natural Fiber and Compatibilizer with PLA we will put it in a die and hit it in 200 degree Celsius . After that we will mix it by using a mixing rod . Then we will keep it at a Natural Temperature after few minutes it will become solidified and the Composite finally made.



Figure 4 - heating the composite material



Figure 5 - Raw specimen



Figure 6 - Final specimen

TENSILE TESTING: Tensile testing is a destructive test process that provides information about the tensile strength, yield strength, and ductility of the metallic material. It measures the force required to break a composite or plastic specimen and the extent to which the specimen stretches or elongates to that breaking point. Tensile testing of composites is generally in the form of basic tension or flat-sandwich tension testing in accordance with standards such as ISO 527-4, ISO 527-5, ASTM D 638, ASTM D 3039, and ASTM C 297. Such tests produce stress-strain diagrams used to determine tensile modulus.



Figure 7 - Tensile testing in UTM machine

APPLICATION OF COMPOSITE MATERIAL:

1. Poly lactic acid composite materials are lightweight , strong and very good fatigue resistance making them suitable for structural applications like aircraft wings.
2. Additionally , PLA is biodegradable and sourced from renewable resources , aligning with the aerospace industry's sustainability goals.

Chapter-4

AIRFOIL NACA 2412: -

Objectives: -

- The parameters to be determined are flow characteristics such as velocity distribution and pressure distribution.
- Determination of performance characteristics such as lift coefficient (C_L) and drag coefficient (C_D).

PROPERTIES OF AIRFOIL NACA 2412: -

- $M=2.0\%$, $P=40.0\%$, $P=12.0\%$

- Chord=1000 mm
- Radius=0 mm
- Thickness=100 mm
- Origin=0 %
- Pitch=0 deg
- It has a maximum camber of 2% located 40% chords (0.4 chords) from the leading edge with a maximum thickness of 12% of the chord.

Properties of Air:

Properties

Density [kg/m ³]	constant	Edit...
	1.225	
Viscosity [kg/(m s)]	constant	Edit...
	1.802e-05	

MODELING AND MESHING: -

- The modelling has been done in ANSYS Workbench.
- NACA-2412 air foil profile was modelled by exporting coordinate data.
- Meshing was done in ANSYS.

Modelling of Air foil:

The air foil model is skilfully design in SOLIDWORKS 2021. To that air foil coordinates are plotted, and the air foil 3D model is created.

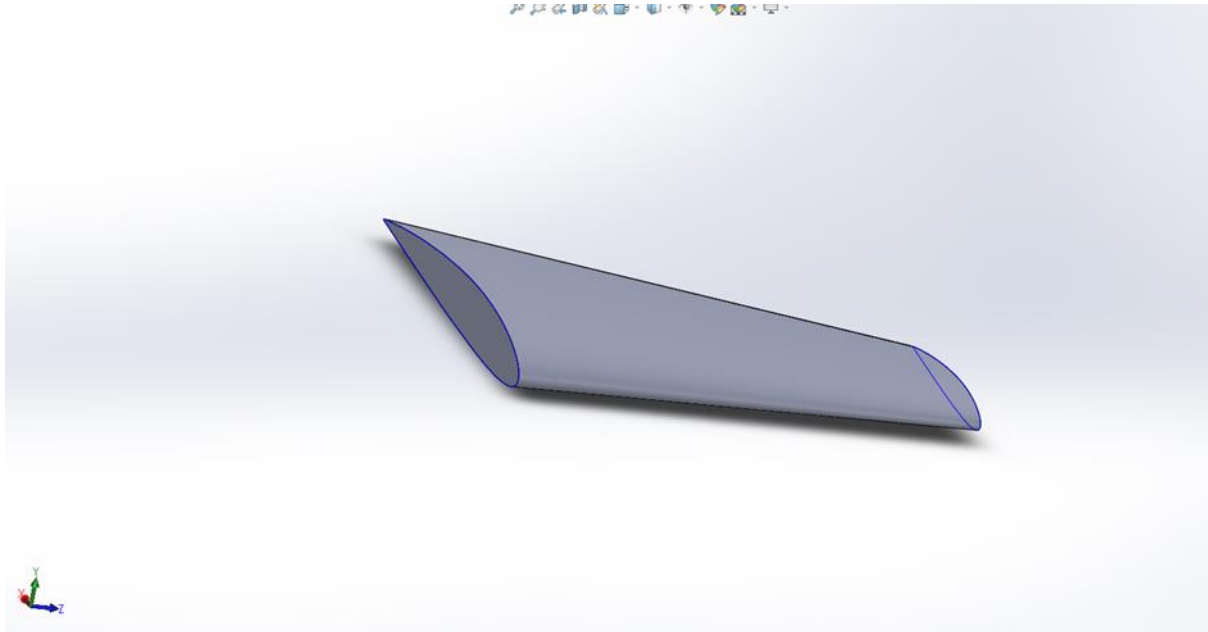
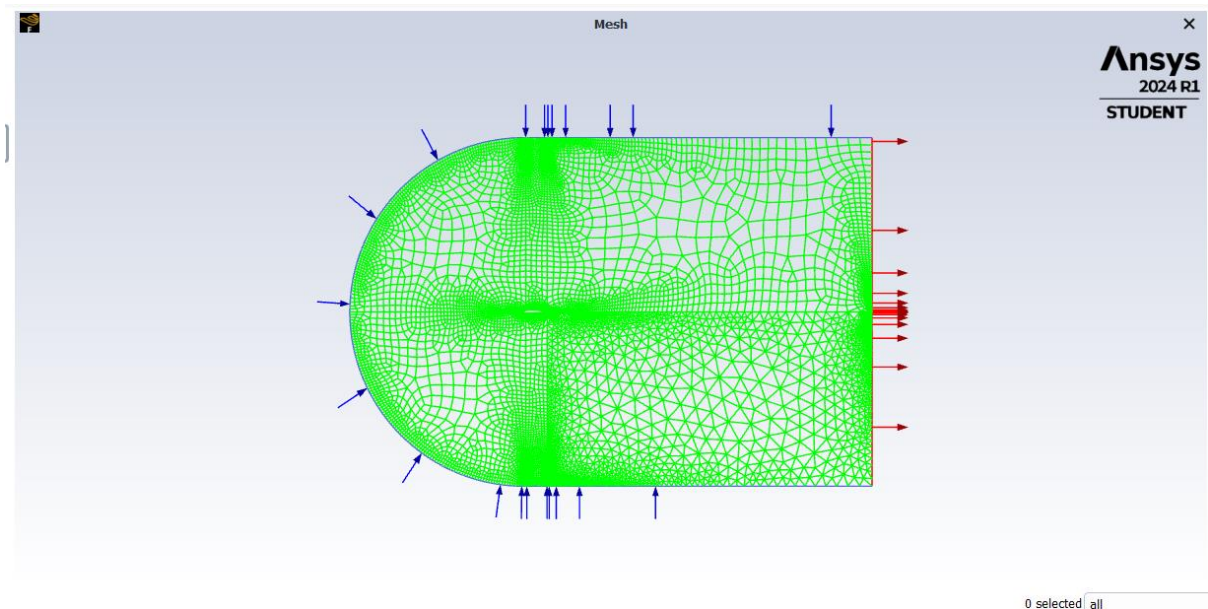


Figure-8 Design of Air foil in SOLIDWORKS

ANSYS is meshing software capable of creating meshed geometries that can be read into Fluent and other analysis software, making a meshed profile is done in 2D. These files are imported in fluent. We have done the meshing of Air foil NACA 2412 and its domain, and then the simulation of flow variables over this control volume is done in the case of 2D of Control Line.



2D Analysis:

The desired meshed can now be read into FLUENT, which will then run the geometry through the numerical analysis. Different angles of attack will be analysed in FLUENT. Air foil and angle of attacks 4, 8, and 12 degrees are analysed.

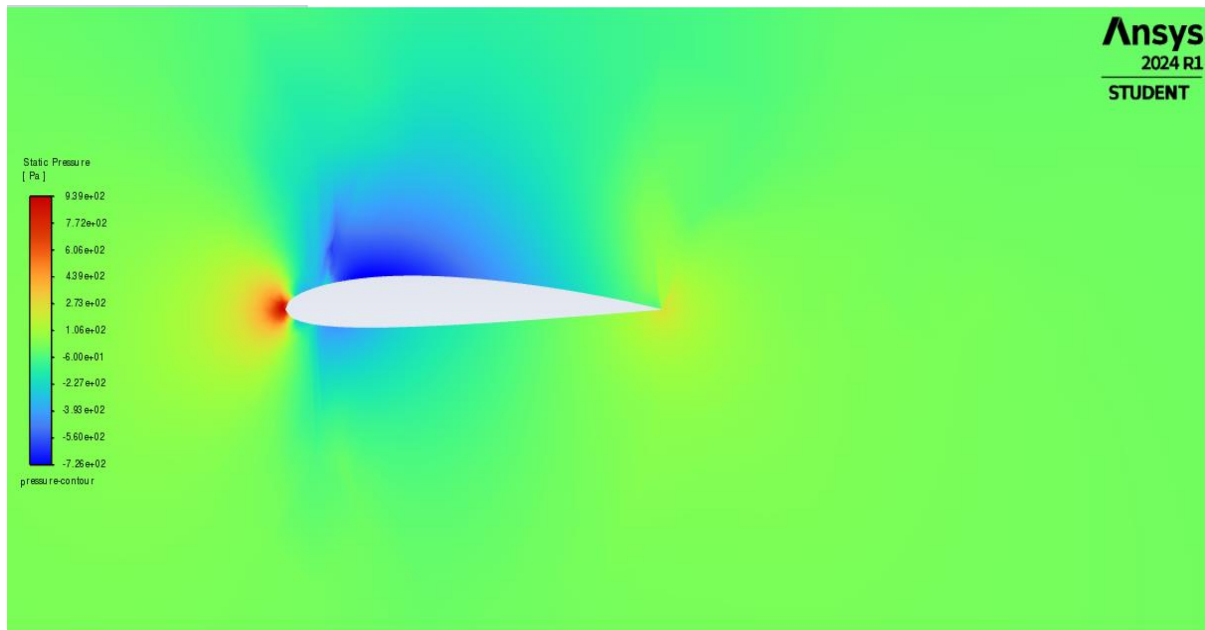


Figure 9- Contour of Pressure

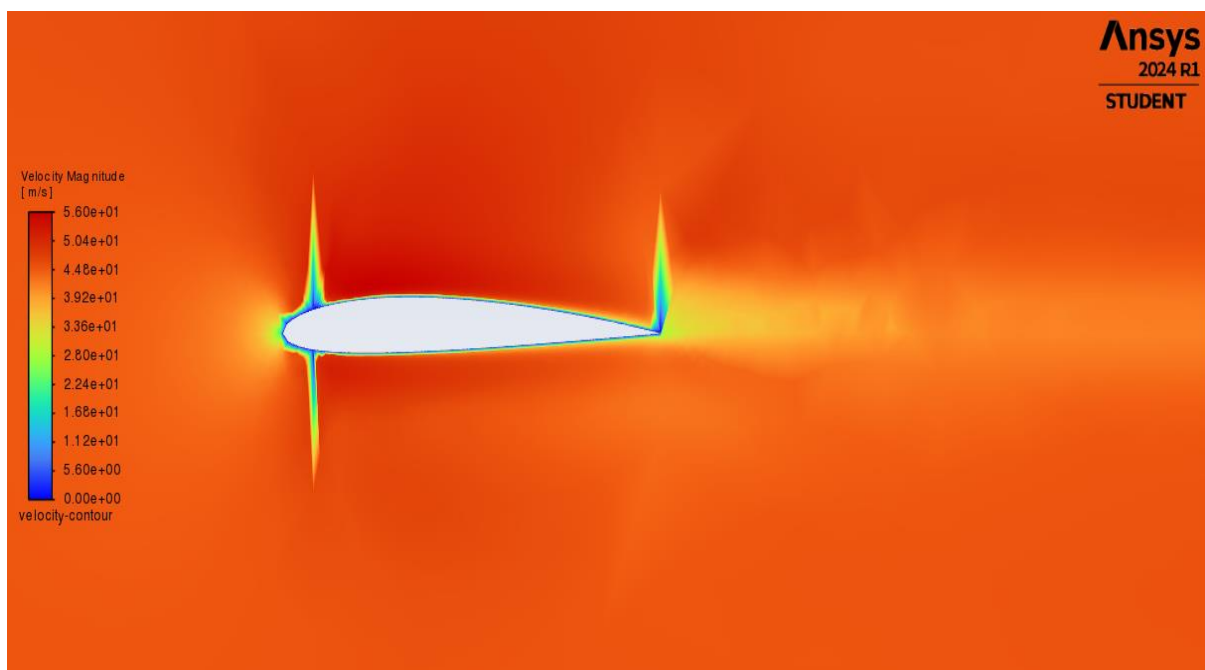


Figure 10- Contour of Velocity

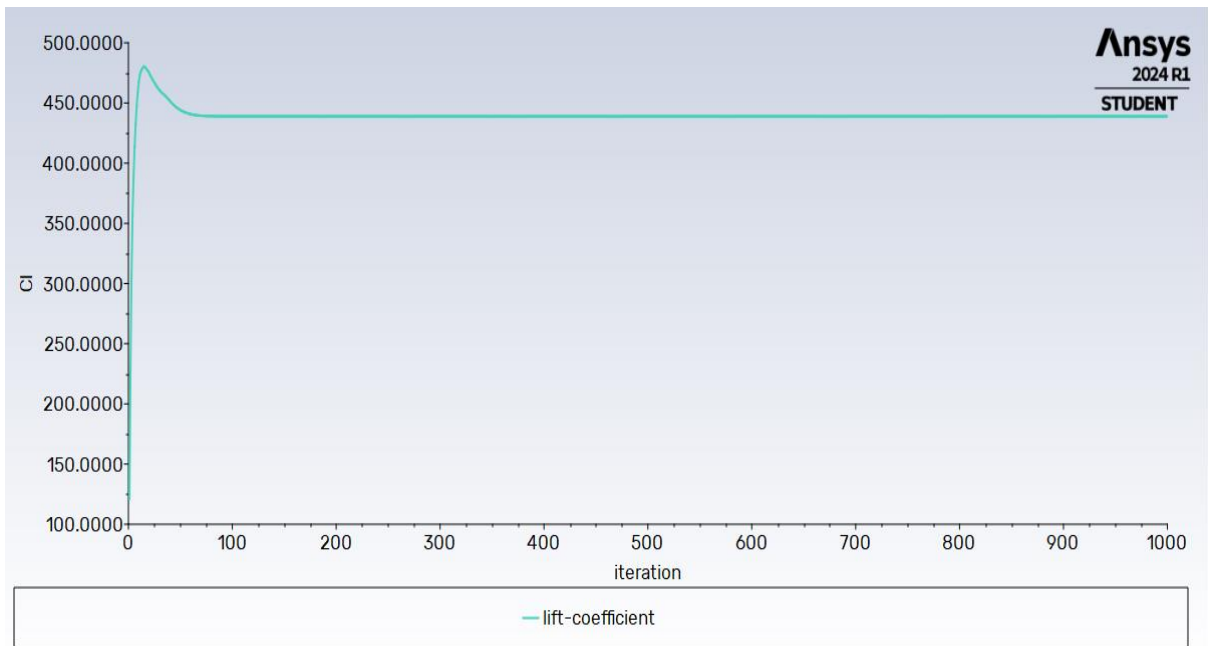


Figure 11- Lift coefficient

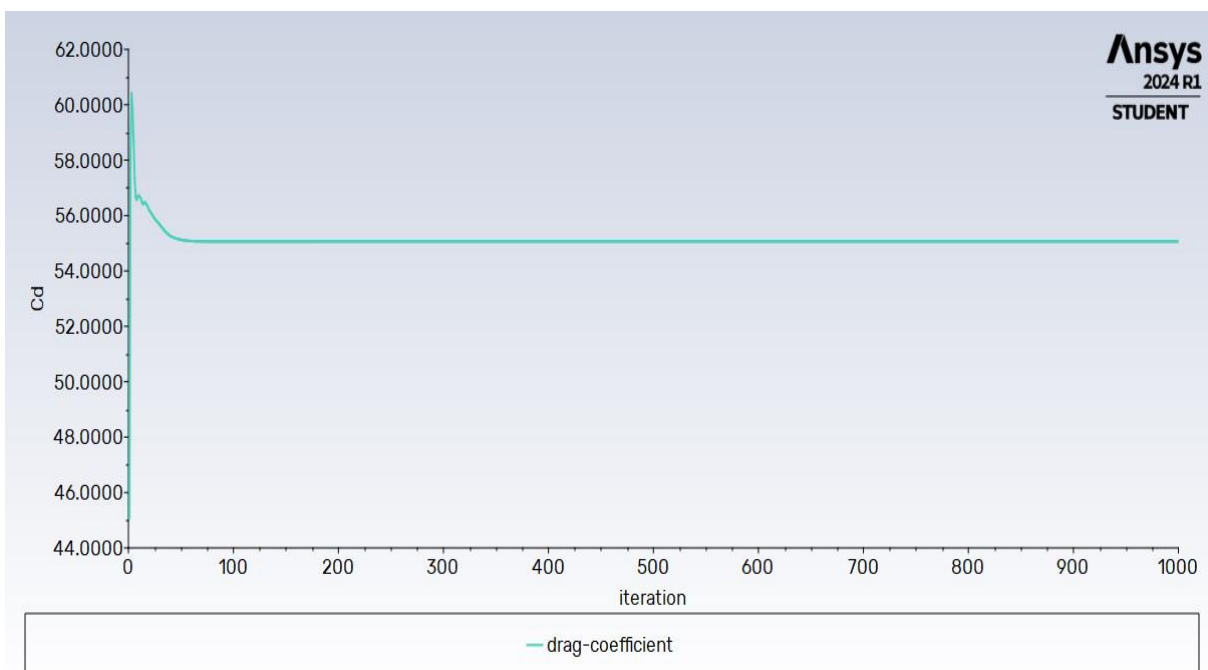


Figure 12- Drag coefficient

Experimental Data Table:

The values of drag and lift coefficient are taken according to this data table.

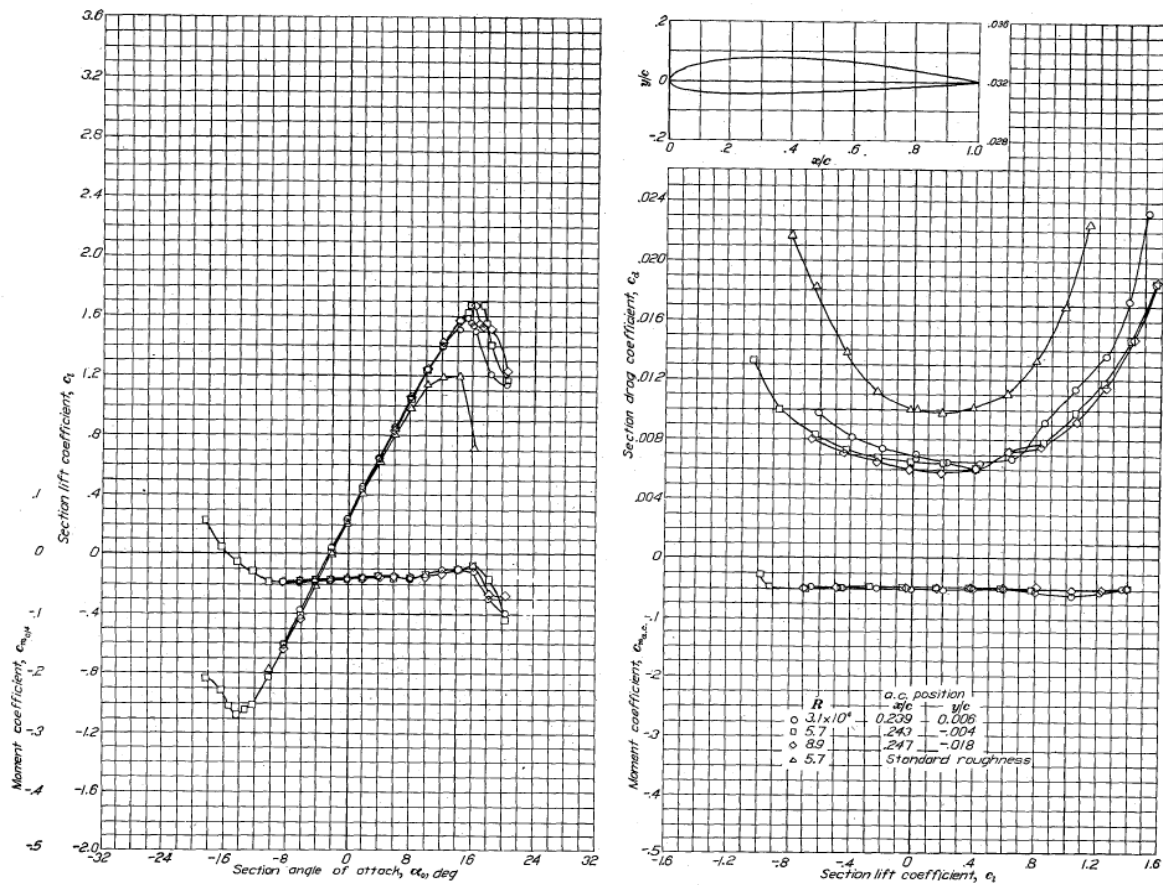


Figure 13- Data Table

Calculation:

$$Re = \frac{\rho u L}{\mu}$$

Density(ρ)=1.225 kg/m³, Chord length(L)=1m, Dynamic Viscosity(μ)=1.802E-5 and Re is Reynolds number.

$$\text{Therefore, } u = \frac{Re \cdot \mu}{L \cdot \rho}$$

$$u = \frac{3.1 \times 10^6 \cdot 1.802 \times 10^{-5}}{1 \cdot 1.225}$$

$$u = 45.6 \text{ m/s}$$

Results:

- At 0-degree angle of attack

The lift coefficient is found to be 0.218 and from experimental it is 0.25 which is only 13% error after 80 iterations.

The drag coefficient is found to be 0.00927 and from experimental data it is 0.0065 which is 30% error after 80 iterations.

- At 8-degree angle of attack,

The lift coefficient is found to be 1.05 which is exactly same as data table therefore it is zero percent error.

The drag coefficient is found to be 0.0014 and from experiment data it is 0.115

References: -

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Chapter-5

CONCLUSION

In conclusion, the integration of compatibilizers in PLA or PLA blends has proven instrumental in enhancing the mechanical properties of green hybrid composites utilized in wind turbine blade applications. Through improved interfacial adhesion and dispersion of constituents, compatibilizers effectively mitigate weaknesses, such as poor interfacial bonding and phase separation, leading to enhanced tensile strength, impact resistance, and fatigue performance. This advancement not only addresses the demand for sustainable materials in wind energy applications but also underscores the importance of innovative materials engineering in advancing renewable energy technologies.

FUTURE SCOPE

Looking ahead, further research could delve into optimizing the formulation of compatibilizers to achieve even greater enhancements in mechanical properties of PLA-based green hybrid composites for wind turbine blade applications. Exploring novel compatibilizer chemistries and processing techniques could lead to superior interfacial adhesion and dispersion, ultimately yielding composites with enhanced strength, stiffness, and durability. Additionally, investigating the long-term performance and environmental sustainability of these enhanced materials would provide valuable insights into their feasibility for widespread adoption in the renewable energy sector. Moreover, exploring the scalability and cost-effectiveness of production methods for these advanced composites could pave the way for their commercialization and widespread deployment in wind energy infrastructure, contributing to a more sustainable and resilient energy future.

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